

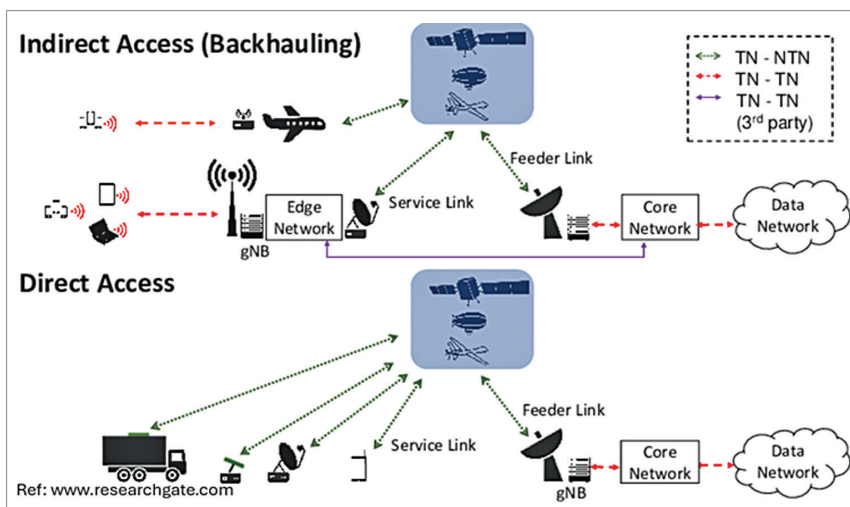


Fig. 2: Two types of last-mile networking (Credit: www.researchgate.com)

transformed the economics of space. Satellite terminals, once large and costly, are now compact, portable, and increasingly affordable.

According to industry projections, the global satellite communication market could exceed \$1 trillion by 2030, with individual national markets, such as India, reaching \$40 billion.

Starlink alone operates over 7000 satellites and aims to surpass 12,000 by 2027. OneWeb, partially backed by Bharti Enterprises, has launched over 600 satellites and is preparing to offer services targeting India and other developing regions.

Such scale ensures that even rural connectivity, traditionally considered unprofitable, becomes viable through hybrid business models combining telecom and space infrastructure.

Technical challenges: The road still ahead

Every innovation brings its hurdles, and 5G-SATCOM convergence is no exception.

• Cost of terminals and service.

While prices are falling, satellite bandwidth and hardware remain costlier than terrestrial alternatives. For rural users, affordability remains key.

• **Weather interference.** High-frequency satellite signals, particularly

in the Ka band, are susceptible to rain fade and atmospheric disturbances, requiring robust adaptive modulation and power control mechanisms.

• **Security and privacy.** SATCOM networks are emerging targets for cyber threats. Encryption, authentication, and cross-domain security protocols are still evolving for large-scale deployment.

• Regulatory coordination.

Spectrum allocation, orbital coordination, and national security considerations demand tight regulatory alignment, especially as multiple commercial players launch global services.

Despite these obstacles, the direction is clear: hybrid connectivity represents the future. Beyond access, Indian engineers are exploring new applications built atop hybrid networks.

One such innovation comes from CinqueNetwork, which is developing personalised communication platforms that turn mobile devices into dynamic, on-demand service hubs. Their concept envisions voice calls, authentication, file sharing, and e-commerce interactions happening “on demand,” without needing multiple installed apps. By building these services on top of hybrid 4G/5G and satellite infrastructures, they aim to make

communication smarter and more adaptable to user needs.

It offers a glimpse into how India’s tech ecosystem is not merely adopting new standards but creating new paradigms around them.

The vision is simple yet profound: a planet where connectivity is ubiquitous, intelligent, and seamless. In that world, a phone will not care whether it is connected to a tower in Delhi, a drone over a flooded region, or a satellite orbiting 1000 kilometres above; it will simply stay connected, automatically choosing the best available link.

Imagine rural healthcare centres performing live teleconsultations, ships relaying real-time weather updates, or farmers accessing IoT-driven irrigation data, all without worrying about network bars. That is the promise of 5G + SATCOM convergence, a world where digital inclusion is no longer bound by geography.

As telecom and space converge, communication is entering its most exciting phase yet. The merging of terrestrial and non-terrestrial networks is not merely an engineering triumph, it is a humanitarian one. It ensures that no classroom, factory, home, or life remains disconnected from opportunity.

From Bengaluru to the high Himalayas, from aircraft cruising over oceans to deep mines beneath the earth, the network of the future is taking shape, everywhere, all at once.

As one technologist aptly summed it up: “The ground is ready. We just have to bring more use cases.” And with 5G and SATCOM working hand in hand, that readiness might soon become reality. **EFY**

This article is based on a tech talk session at IEW 2025, Bengaluru, by Prasenjit Das, Co-Founder, CinqueNetwork. It was transcribed and curated by Akanksha Sondhi Gaur, Senior Technical Journalist at EFY.